

CarbonSmart Program

Program Name: CarbonSmart

Objective: To fund Ohio's transition to a renewable energy economy through a carbon emissions tax, which finances two distinct grant programs:

1. A grant for college students pursuing renewable energy-related majors.
 2. A grant for small businesses starting or expanding renewable energy and sustainability-focused operations.
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1. CarbonSmart Overview

CarbonSmart aims to reduce Ohio's carbon footprint by taxing large carbon-emitting industries and reinvesting those funds into the state's renewable energy future. This initiative will create educational opportunities for students and stimulate economic development by supporting small businesses committed to renewable energy and sustainability.

2. Carbon Emissions Tax Structure

- **Tax Rate:** The program will introduce a tax of **\$10 per metric ton of CO₂e** (carbon dioxide equivalent).
- **Threshold:** The tax will apply to industries emitting more than **100,000 metric tons of CO₂e per year**. This targets large-scale emitters while sparing small businesses or mid-sized emitters from the tax burden.
- **Targeted Emitters:**
 - **Power plants:** Coal and natural gas facilities.
 - **Manufacturing:** Steel, automotive, and chemical plants.
 - **Transportation & logistics:** Large-scale freight companies, rail, and logistics hubs.
 - **Agriculture:** Large agribusinesses involved in methane and CO₂-intensive operations.

3. Revenue Projections and Allocation

Given Ohio's emissions profile, the carbon tax is projected to generate **\$100 million to \$300 million annually** during the early stages. The tax rate and scope can gradually increase over time, but this initial range reflects a scaled-back, manageable approach to launching the program.

Revenue Breakdown:

- **50% for College Student Grants**
 - **50% for Small Business Grants**
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4. Program Components

Part 1: College Student Grant Program

This program aims to develop a workforce skilled in renewable energy and sustainability. It supports Ohio residents enrolled in accredited institutions with majors that align with the state's clean energy transition.

- **Eligibility:**
 - Must be an Ohio resident.
 - Enrolled in a renewable energy-related major such as:
 - Renewable Energy Engineering
 - Environmental Science
 - Sustainable Agriculture
 - Energy Economics
 - Environmental Policy
 - Must demonstrate financial need or academic merit.
- **Grant Amounts:**
 - Awards ranging from **\$2,500 to \$10,000 per year**, based on financial need, academic performance, and program duration.
- **Application Process:**
 - Submit proof of enrollment in a qualifying major.
 - Provide a project proposal or essay explaining the student's interest in renewable energy and their career goals.
 - Optional: Letters of recommendation from professors or industry mentors.
- **Post-Graduation Incentives:** Additional support for graduates who work in Ohio's renewable energy or sustainability sectors after graduation.

Part 2: Small Business Grant Program

The small business grant program aims to support entrepreneurs and companies developing innovative renewable energy or sustainability projects. It targets businesses with the potential to reduce Ohio's carbon footprint and contribute to the state's green economy.

- **Eligibility:**
 - Must be a small business registered in Ohio.
 - Must focus on renewable energy production, energy efficiency, or sustainability, such as:

- Solar energy installation
 - Wind energy solutions
 - Energy-efficient building materials
 - EV infrastructure and services
 - Waste-to-energy technologies
 - Sustainable agriculture practices
 - Businesses located in underserved or high-emission regions will receive priority.
 - **Grant Amounts:**
 - Grants range from **\$10,000 to \$50,000**, depending on the size and scope of the business and its environmental impact.
 - Funding may be used for:
 - Product development or innovation.
 - Scaling up operations.
 - Marketing, hiring, or capital improvements related to clean energy.
 - **Application Process:**
 - Submit a detailed business plan, including projected environmental impact, market potential, and financial sustainability.
 - Provide documentation of matching funds or external investment for larger grants.
 - Businesses will undergo a review process to assess their viability and alignment with program goals.
 - **Support Services:**
 - Grant recipients will have access to technical, financial, and legal advisory services to help scale their businesses and ensure long-term success.
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5. Implementation Strategy

Step 1: Legislative and Regulatory Framework

- **CarbonSmart Legislation:** Introduce and pass legislation in Ohio's state legislature to authorize the carbon emissions tax and define program parameters.
- **Oversight:** Establish a regulatory body or appoint the Ohio Environmental Protection Agency (EPA) to manage emissions monitoring and tax collection.
- **Emissions Monitoring:** Leverage existing emissions reporting infrastructure (such as the EPA's Greenhouse Gas Reporting Program) to streamline the process for businesses.

Step 2: Partnerships

- **Universities & Colleges:** Collaborate with Ohio's higher education institutions to promote the student grant program and attract qualified applicants.

- **Industry Partnerships:** Work with renewable energy industry groups to promote the small business grants and help ensure that businesses have the resources and networks to succeed.
- **Local Governments:** Partner with local governments to assist in program promotion, particularly in underserved areas, and create localized clean energy projects.

Step 3: Public Awareness and Outreach

- **Marketing Campaign:** Launch a public campaign to raise awareness about the program's benefits for reducing emissions and promoting economic growth.
 - **Workshops & Webinars:** Host events to explain the grant application processes and educate businesses on emission reduction strategies.
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6. Monitoring and Evaluation

Annual Reviews:

- Evaluate the revenue generated, the distribution of grants, and the overall program success. Adjust the tax rate or revenue allocation as needed based on program performance.
- Conduct emissions audits to ensure compliance with the tax and monitor whether businesses are reducing their carbon footprints.

Success Metrics:

- **For College Student Grants:**
 - Number of students funded.
 - Post-graduation employment in renewable energy sectors in Ohio.
 - Contributions to renewable energy innovation.
 - **For Small Business Grants:**
 - Number of businesses started or expanded.
 - Jobs created in renewable energy and sustainability fields.
 - Measurable reductions in carbon emissions or increased adoption of renewable energy solutions.
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7. Long-Term Expansion and Adjustments

Scaling the Tax:

- After a few years, consider gradually increasing the carbon tax rate (e.g., raising it by \$5 per metric ton every 2-5 years) and lowering the emissions threshold to include more businesses, while continuing to incentivize emission reductions.

Adjusting Grant Programs:

- Based on demand and success, expand the grant programs to include more majors, businesses, or additional forms of renewable energy innovation.
 - Introduce new incentives for businesses that achieve substantial reductions in their carbon emissions.
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8. Challenges and Mitigations

- **Industry Pushback:** Offering performance-based tax credits for emission reductions and carbon capture projects can mitigate industry resistance.
 - **Economic Sensitivity:** By starting with a modest tax and higher threshold, CarbonSmart minimizes initial economic disruption while allowing businesses time to adapt.
 - **Administrative Burden:** The program can leverage existing emissions reporting mechanisms, reducing the complexity of implementation.
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9. Conclusion

The **CarbonSmart** program represents a forward-looking initiative that will position Ohio as a leader in renewable energy and sustainability. By implementing a modest carbon tax and reinvesting the revenue into educational and entrepreneurial grants, Ohio can reduce emissions, support innovation, and create a stronger, greener economy for future generations.